



# Affective Image Classification using Features Inspired by Psychology and Art Theory

Jana Machajdik & Allan Hanbury, 2010

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# Overview

## Main objective

- **Emotional Semantic Image Retrieval (ESIR):** Classify images into their perceived emotional content
- Find features in an image that correspond to a specific emotional content

## Method

- Use concepts from psychology and art to define features that are potentially connected to emotional expression (e.g. color red to anger)
- Classification into **eight emotional categories:** Amusement, Awe, Contentment, Excitement, Anger, Disgust, Fear and Sad
- Determine the most relevant features

# Background and importance

## Motivation

- Huge and constantly growing image databases on the internet
- Need for **categorization based on emotional content** (e.g. journalism, AI generated images)
- Most applications lack the ability to incorporate the affective level
- Goal to **mimic human decisions**. Step towards training machines to human emotions.

## Preliminary work

- Few publications in affective content analysis
- There is advanced work but it usually has
  - too common and generic features or
  - chosen emotional categories are too specific to the studies

# Methodology

- **Preprocessing**

- Image resizing and cropping
- Color space conversion (RGB to cylindrical)
- Waterfall segmentation algorithm

- **Feature extraction**

- Define features
- Compute features for each image
- 114 total features

# Preprocessing

- **Resize images**
  - To 200,000 pixels (maintaining aspect ratio)
  - Gain computational efficiency
- **Remove borders**
  - Hough transform + Canny edge detector
  - Crop single-color borders
- **Convert color space**
  - RGB to cylindrical coordinate color space
  - Intuitive definition of Hue (H), Saturation (S), and Brightness (Y) channels
- **Image segmentation**
  - Waterfall segmentation to get contiguous regions within images
  - Analyze spatial composition

# Features

- **Feature categories**

- Psychology-based features
  - ▶ Saturation and brightness related to emotional dimension
- Art theory-based features
  - ▶ Relate color combinations to induced emotional effects
- Standard features
  - ▶ Proven utility in image retrieval and classification tasks

- **Feature groups implemented**

- Color
- Texture
- Composition
- Content

# Color features

- **Saturation and brightness**
  - pleasure, arousal and dominance
- **Hue** (tone)
- **Colorfulness**
  - with Earth Mover's Distance (EMD)
- **Color names** (number of pixels per color)
- **Itten contrasts** and color accordances
- **Wang's histogram**
  - emotional impact of color

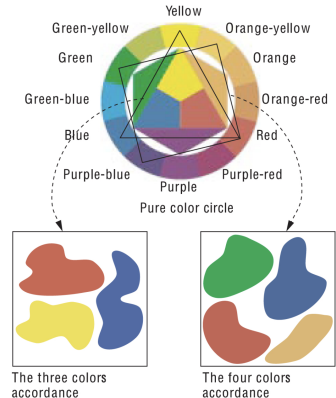


Figure: Color accordances.

# Texture features

- **Wavelet-based** features
  - Measure spatial smoothness/graininess
  - Compute 9 wavelet features (H, S, Y) over 3 levels
- **Tamura** texture features
  - Coarseness, contrast, and directionality
- **Gray-Level Co-occurrence Matrix (GLCM)**
  - Compute contrast, correlation, energy, and homogeneity

# Composition features

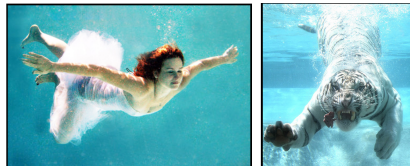
- **Level of Detail**
  - Impact of image detail
- **Low Depth of Field (DOF)**
  - Focus on the main subject
  - Blurring of the background
- **Dynamics**
  - Emotional effects of lines
  - Different lines convey different emotions
- **Rule of Thirds**
  - Divide the image into thirds, positioning the main object



**Figure:** Rule of Thirds applied to two images.

# Content features

Semantic content has the greatest impact on emotional influence.



**Figure:** Two similar images with different contents.

- **Human faces**

- Faces draw attention
- Detect frontal faces using Viola and Jones' algorithm
- Count faces and consider the size of the largest face

- **Skin detection**

- Detect “artistic nudes”
- Measure skin area and its proportion to detected face size

# Experiment

**Emotional Output Categories:** Amusement, Awe, Contentment, Excitement, Anger, Disgust, Fear, Sadness (see psychological study on affective images).

## Setup

- Rule: “one category against all”
- Optimize for the true positive rate per class averaged over the positive and negative classes

## Experiment - Testing and training data set

- A set of 394 images from **The International Affective Picture System (IAPS)**
  - Used in emotion research
  - Documentary natural color photos depicting objects
- A set of 807 **artistic photographs** downloaded from an art sharing site
  - Emotion determined by the artist
  - Conscious manipulation of the image composition, lighting, colors, etc
- A set of 228 peer rated **abstract paintings**
  - Combinations of color and texture
  - Each image rated about 14 times
- A combined set of all of the above

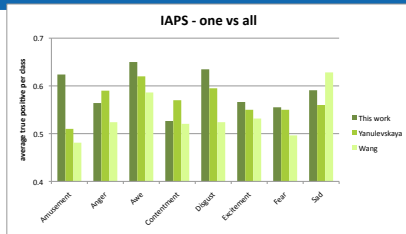
# Classification performance results

- **IAPS:**

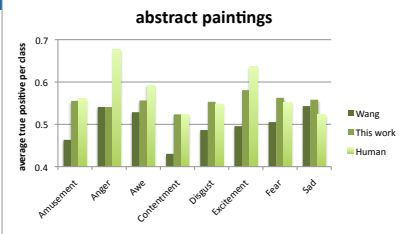
- Best feature set is dependent on both the category and the data set
- Occurrence and size of human faces is the strongest feature for the Amusement category
- Categories are strongly content related

- **Artistic sets**

- No strong connection between faces and categories
- Colors are more important for photos (features from art theories (Itten colors, Wang Wei-ning histograms) are more common)
  - ▶ Itten features are selected for the Amusement and Excitement classes
  - ▶ Wang Wei-ning et al. features are selected most for Awe and Disgust classes (also selected for the Amusement and Excitement classes)



(a) Performance for IAPS.

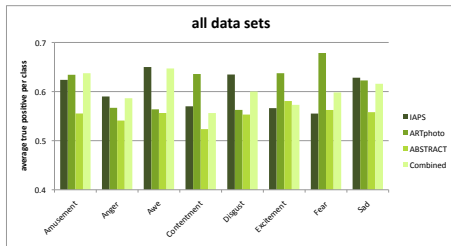


(b) Performance of abstract paintings set.

**Figure:** Classification performance taking best features for each category against the best features from previous works

- This work's feature sets outperform results by Yanulevskaya and Wang Wei-ning features in case of IAPS set for five categories
- This work's feature set and humans obtained similar results in the abstract paintings data set (not in the anger class)

# Classification performance for all image sets



**Figure:** Classification performance for all image sets. Results are from the best feature selections

- No clear difference between IAPS and the artistic photo image sets
- Abstract paintings data set has the worst performance

# Discussion and Conclusion

The algorithm achieved better results than the state of the art.

Critics: Cultural differences in how features and emotion are perceived could lead to false conclusions and discrimination.

It could be improved through

- New features (emotional expression, identify symbols (e.g. hearts, crosses))
- New Classification (distribution over the categories)
- More reliable ground truth (take larger samples of people)
- Different approach (drop consensus and learn individual preferences)
- Further research in psychology of affective images

Thank you!  
Questions?

# References

Jana Machajdik & Allan Hanbury, *Affective Image Classification using Features Inspired by Psychology and Art Theory*, 18th ACM international conference on Multimedia, 2010

[Link to the pdf version](#)

| Category    | Short Name                          | #  | Short Description                                                                                                                                                                                                                                                                                                               |
|-------------|-------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| color       | <i>Saturation, Brightness</i>       | 2  | mean saturation and brightness                                                                                                                                                                                                                                                                                                  |
|             | <i>Pleasure, Arousal, Dominance</i> | 3  | approx. emotional coordinates based on brightness and saturation                                                                                                                                                                                                                                                                |
|             | <i>Hue</i>                          | 4  | vector based mean hue and angular dispersion, saturation weighted and without saturation                                                                                                                                                                                                                                        |
|             | <i>Colorfulness</i>                 | 1  | colorfulness measure based on EMD                                                                                                                                                                                                                                                                                               |
|             | <i>Color Names</i>                  | 11 | amount of black, blue, brown, green, gray, orange, pink, purple, red, white, yellow                                                                                                                                                                                                                                             |
|             | <i>Itten</i>                        | 20 | average <i>contrast of brightness, contrast of saturation, contrast of hue, contrast of complements, contrast of warmth, harmony, hue count, hue spread, area of warm, area of cold</i> ,...and the maximum of each features (histograms) by Wang Wei-ning et al. [31] ( <i>factors 1 (10), factor 2 (7) and factor 3 (2)</i> ) |
|             | <i>Wang</i>                         | 19 | based on Wang features: <i>area of very dark, area of dark, area of middle, area of...light, very light, high saturation, middle saturation, low saturation, warm, cold</i>                                                                                                                                                     |
| texture     | <i>Area statistics</i>              | 10 |                                                                                                                                                                                                                                                                                                                                 |
|             | <i>Tamura</i>                       | 3  | features by Tamura et al [25]: <i>coarseness, contrast, directionality</i>                                                                                                                                                                                                                                                      |
|             | <i>Wavelet textures</i>             | 12 | wavelet textures for each channel (Hue, Saturation, Brightness) and each level (1-3), sum of all levels for each channel                                                                                                                                                                                                        |
| composition | <i>GLCM-features</i>                | 12 | features based on the GLCM: <i>contrast, correlation, energy, homogeneity</i> for Hue, Saturation and Brightness channel                                                                                                                                                                                                        |
|             | <i>Level of Detail</i>              | 1  | number of segments after waterfall segmentation                                                                                                                                                                                                                                                                                 |
|             | <i>Low Depth of Field (DOF)</i>     | 3  | low depth of field indicator; ratio of wavelet coefficients of inner rectangle vs. whole image (for Hue, Saturation and Brightness channel)                                                                                                                                                                                     |
|             | <i>Dynamics</i>                     | 6  | Line slopes: static, dynamic (absolute and relative), lengths of static lines, lengths of dynamic lines                                                                                                                                                                                                                         |
| content     | <i>Rule of Thirds</i>               | 3  | mean saturation, brightness and hue of the inner rectangle                                                                                                                                                                                                                                                                      |
|             | <i>Faces</i>                        | 2  | number of frontal faces, relative size of the biggest face                                                                                                                                                                                                                                                                      |
|             | <i>Skin</i>                         | 2  | number of skin pixels, relative amount of skin with respect to the size of faces                                                                                                                                                                                                                                                |

Figure: Features summary.

|                    | Amusement | Anger | Awe | Contentment | Disgust | Excitement | Fear | Sad | sum  |
|--------------------|-----------|-------|-----|-------------|---------|------------|------|-----|------|
| IAPS               | 37        | 8     | 54  | 63          | 74      | 55         | 42   | 61  | 394  |
| Art photo          | 101       | 77    | 103 | 70          | 70      | 105        | 115  | 166 | 807  |
| Abstract paintings | 25        | 3     | 15  | 63          | 18      | 36         | 36   | 32  | 228  |
| Combined           | 163       | 88    | 172 | 196         | 162     | 196        | 193  | 259 | 1429 |

Figure: Number of images per data set and emotion category.

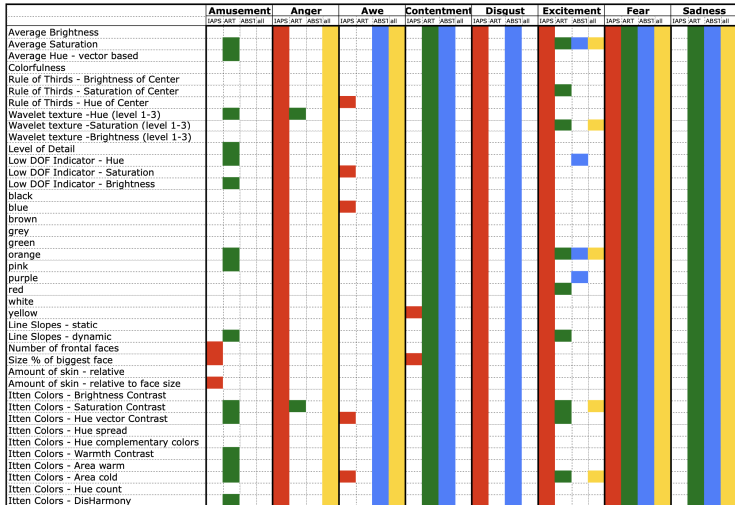


Figure: Detailed results (1/2).

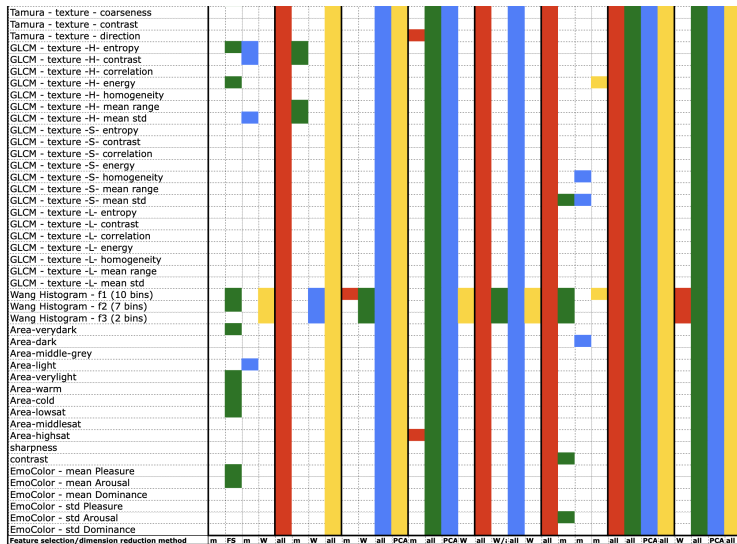


Figure: Detailed results (2/2).